

TEZPUR UNIVERSITY

Test I, Autumn 2014

C0 504 Natural Language Processing

Marks : 25

Time : 30 minutes

The figures in the right-hand margin indicate marks for the individual question.

1. From the NLP perspective what is a *language*? What are some differences between a programming language and a natural language? 2 + 3
2. Write briefly about representation of speech for computational purposes. How is Nyquist-Shannon theorem relevant in this? 3 + 2
3. (a) Unicode does not enforce any representation of the codes assigned to characters of a script. Explain. 4
(b) What is UTF-8? 4
4. What is a non-deterministic finite automata? What is its use? 3 + 2
5. Write the Perl regular expression to recognize strings that start with *The* and end with a dot. 2